Effects, Not Eigenstates: The Born Rule as a Theorem, and the Boundary Where This Framework's Account of Measurement Stops

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A short companion, in the register of *The Permission Slip*: one cited theorem, one honest boundary, kept apart. It proves nothing about outcome selection; it establishes precisely how much of ordinary quantum statistics this framework gets for free, and names exactly where that stops.

Revision 4.2. Two locator corrections. The Hamhalter citation named the wrong chapter — chs. 5–6 ("Generalized Gleason Theorem," "Basic Principles of Quantum Measure Theory"), not ch. 3, which the book's own table of contents and its chapter-5 abstract confirm treats only the classical Hilbert-space case; the correction adds a pointer to the book's own Yosida–Hewitt decomposition, the exact theorem-level home of the normal-versus-singular distinction this document uses informally. Second, "established directly in Keystone §11" overstated what that section does — it invokes ω_β's faithfulness as a premise, not proves it; the repair states the correct, standard reason (every KMS state is faithful on the von Neumann algebra it generates, via the KMS analyticity condition, Bratteli–Robinson) and adjusts the verb accordingly. Two clause-level phrasing fixes also applied: "in the appropriate topology" moved from the finite-additivity-on-effects clause, where it was vacuous, to the complete-additivity-on-projections clause, where a topology is actually needed; and the finite-additivity clause now says "finite families" explicitly rather than leaving it implicit.

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1. Why the textbook picture is unavailable here, and it is a theorem, not a stance

[FACT, cited] This project's operational algebra is the type III_1 factor, established independently from the octonionic side and now, through *The Keystone*, traceable to the seed with a precisely located remaining ansatz. Type III factors admit no minimal projections and no pure normal states — a standard structural fact, not an assumption this document introduces.

[FACT, cited] The corpus already states the consequence precisely, and this document adopts the statement rather than re-deriving it — from *The Two Halves of One Product: The Jordan/Malcev Split of the Octonions and the Measurement Channel* (§4): "in the type III_1 setting there is nothing for the projection picture to refer to — no minimal projections and no pure normal states exist… Events are irreducibly coarse; collapse-to-pure-outcome has no referent." Measurement, evaluated through a state applied to an observable rather than through projection onto an eigenstate, is correspondingly the only formalization available — not a preference, a consequence of the algebra's own type. A second corpus document, *The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering* (§1), independently confirms the same fact is what forces its own reconstruction of measurement: "type III algebras have no minimal projections and no pure normal states, so the textbook projection postulate is mathematically unavailable for local measurements — there is nothing to collapse onto."

[SCOPE] This closes the textbook picture unconditionally. It does not, by itself, tell you what replaces it, nor whether *any* rule assigning probabilities to events is forced rather than freely chosen. §2 addresses the second question directly.

[PROP, with an [INTERPRETATION] reading — from review correspondence] Why collapse could ever have seemed available is itself explainable, and worth one paragraph. In type I — ordinary quantum mechanics on $B(H)$ — two logically distinct levels coincide: a maximally sharp state is literally a minimal projection, one object playing both roles, and the projection postulate lives entirely on that coincidence — it asks the post-measurement state to be an element of the algebra. In type III the coincidence is impossible in principle, not merely at the sharp end: with no trace, no state whatsoever — pure or mixed — can be written as trace-pairing against an algebra element, so states and events are permanently different kinds of thing, and the postulate loses not its truth but its referent. What survives is conditioning: measurement genuinely refines the state, but refinement has no terminus — no purity to arrive at — and it is an operation on the assignment of values, not on anything in the arena: no observable inside the algebra registers it, and it has no spatial address. From within the emergent description, an update with no location necessarily presents as an unlocatable, instantaneous jump — the appearance of collapse, without a collapse process. This explains where the jump is not and why it looks as it does; it says nothing about why this outcome occurs in this run, which remains §5's boundary exactly.

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2. The theorem that fills the gap, stated precisely

[FACT, cited] (Christensen, 1982.) For a von Neumann algebra with no direct summand of type I_2 and no direct summand of type II_1, every finitely additive probability measure on its lattice of projections is the restriction of a state. (Christensen, E., "Measures on projections and physical states," *Commun. Math. Phys.* 86 (1982), 529–538.) This is the case used below and is sufficient on its own for what follows; the full removal of the type-II_1 exclusion — the general statement sometimes cited as Gleason–Christensen–Yeadon, valid for *any* von Neumann algebra without a type I_2 summand — is the completion of the Mackey–Gleason problem by Bunce and Wright (Bunce, L. J. & Wright, J. D. M., "The Mackey–Gleason Problem," *Bull. Amer. Math. Soc.* 26 (1992), 288–293), building on Yeadon's resolution of the type II_1 case specifically (Yeadon, F. J., "Measures on projections in W^*-algebras of type II_1," *Bull. London Math. Soc.* 15 (1983), 139–145). This document does not need the general form and cites Christensen's 1982 result alone for §2's application.

[FACT] This applies to this project's algebra directly, not by an extension or a special case. The operational algebra is a *factor* — trivial center, proven independently through the centralizer argument — and a factor has no summand of any kind besides itself; being specifically type III_1, it is neither type I_2 nor type II_1. Christensen's 1982 hypothesis is satisfied outright.

[FACT, cited] (Busch, 2003.) The theorem extends from projections to *effects* — the positive operators bounded between 0 and the identity that occur as elements of a positive-operator-valued measure (POVM), the standard generalization of a sharp observable to an unsharp one. Any generalized probability measure on the effects of a Hilbert-space algebra is the restriction of a state given by a density operator, proved by a route independent of Gleason's original and valid without the type-I_2 exclusion in the effect-valued setting specifically. (Busch, P., "Quantum States and Generalized Observables: A Simple Proof of Gleason's Theorem," *Phys. Rev. Lett.* 91 (2003), 120403.) Busch's own statement concludes a density operator, as stated for $B(H)$; the argument itself is effect-level additivity, nearly linearity already, which is exactly why it avoids the type-I_2 obstruction and transfers verbatim to any von Neumann algebra — Busch's own argument carries the effect-level step; the general Gleason-type literature on von Neumann algebras, cited below for the completely-additive/normality refinement §3 needs, is a separate anchor for a separate step — yielding a state rather than a density operator in the type III setting relevant here, since type III factors admit no density operators at all — there is no trace to define them. This is the form directly relevant here, since — §4 below — nothing in this framework's own measurement construction produces a sharp projection at all.

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3. What this delivers: the Born rule as a theorem, not a postulate

[PROP] Put together, §§1–2 give a specific, checkable claim, not an interpretive gloss. Assume only that measurement outcomes correspond to effects on this algebra, and that probability assignment is finitely additive on effects (over *finite* families with $\sum_i E_i \leq \mathbb{1}$) together with completely additive on the projections among them — the two-part structural content that actually forces the conclusion below, stated in full precision in the fact that follows. The Gleason–Busch theorem then *forces* that assignment to come from a genuine normal state via the standard evaluation formula: no other rule meeting this structural content exists, on an algebra of this type, that fails to reduce to $\rho(E)$ for some normal state ρ evaluated directly on the effect — the state-on-algebra pairing that remains well-defined in type III, where $\mathrm{Tr}[\rho E]$-style notation has no referent, exactly as this document's own §1–2 already establish. This is the precise sense in which this framework can claim the Born rule as *derived from its own premises* rather than independently postulated: the premises are the algebra's own type and the structural requirement of additivity, both already established elsewhere in this corpus, not new assumptions introduced for this purpose.

[FACT, corrected] The additivity premise needs to be read precisely, not left at the informal level "additive on orthogonal effects" might suggest — which, on inspection, is imprecise on two counts, both repaired here. *First, the object level:* effects are general positive operators between 0 and $\mathbb{1}$, not projections, and "orthogonal" is not the condition that governs them — two effects can satisfy $E + F \leq \mathbb{1}$ without $EF = 0$, and neither condition is what the argument actually needs. The precise premise splits into its two working parts: finite additivity on effects (Busch's linearity input, over finite families with $\sum_i E_i \leq \mathbb{1}$ — a finite sum, needing no topology) plus *complete* additivity on the projection sublattice specifically (the normality input, over arbitrary, possibly infinite orthogonal families, in the σ-weak topology, below). *Second, the additivity strength itself:* Christensen's theorem (§2), as cited, needs only *finite* additivity on projections to conclude that some state extends the given measure — but finite additivity alone does not guarantee that state is *normal*: normality of a state on a von Neumann algebra is equivalent to complete additivity, i.e. additivity over *arbitrary* orthogonal families of projections, not merely countable ones, as a fully general fact. (Countable additivity suffices to guarantee normality specifically on a σ-finite algebra, where every orthogonal family of nonzero projections is automatically countable, so the two strengths coincide; this project's own operational algebra carries the faithful normal state ω_β — faithful as every KMS state is on the von Neumann algebra it generates, a standard consequence of the KMS analyticity condition making the GNS vector separating (Bratteli–Robinson, *Operator Algebras and Quantum Statistical Mechanics II*), invoked rather than re-derived in *Keystone* §11 — and a von Neumann algebra admitting a faithful normal state is σ-finite by that fact alone, so countable additivity is sufficient *here*, though this is a consequence of a corpus fact, not a general equivalence, and is not invoked as one.) The generalized Gleason theorem this needs — linear extension of bounded, completely additive measures on a projection lattice to the full von Neumann algebra — is precisely Hamhalter's own subject (*Quantum Measure Theory*, Fundamental Theories of Physics vol. 134, Kluwer/Springer, 2003, chs. 5–6, "Generalized Gleason Theorem" and "Basic Principles of Quantum Measure Theory" — not ch. 3, which treats only the classical Hilbert-space case; the book's own ch. 6 additionally carries the Yosida–Hewitt decomposition of quantum measures, the theorem-level home of the normal-versus-singular split this paragraph otherwise describes only informally), cited here where it actually applies, corrected from an earlier draft's mis-scoped placement against the effect-level Busch transfer in §2, and from this same revision's own first attempt at the chapter locator. A merely finitely-additive measure can in principle be extended by a singular state — mathematically permitted, but physically pathological: singular states are precisely those invisible to every density-matrix-like approximation, not arising from any physically preparable ensemble, which is the relevant sense in which "the Born rule" fails to apply to them. For the physical reading this document intends, the additivity premise therefore needs to be complete additivity on projections, stated as such — a second, distinct premise from the one flagged in the scope note below, made explicit here on the same honesty footing rather than left folded silently into "additive."

[SCOPE] This is a narrower and more honest claim than "the Born rule is derived here" read without qualification. The Born-rule-derivation literature is contested — the additivity assumption itself (now split and stated precisely above: finite additivity on effects, complete additivity on projections) has been challenged as doing more work than it appears to, and this document takes no position on that debate beyond what is needed here. What is being claimed is exactly this: *given* this framework's own algebra and *given* the standard structural axioms of quantum probability theory — additivity read at the strength that actually delivers normality — the evaluation rule is forced, not chosen. Nothing broader is asserted.

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4. Where the concrete statistics actually come from

[FACT, cited] This framework already has a working, non-projective measurement construction, not merely the abstract permission §§1–3 establish. The Fewster–Verch scheme (Fewster, C. J. & Verch, R., "Quantum fields and local measurements," *Commun. Math. Phys.* 378 (2020), 851–889) — a probe field dynamically coupled to the system in a bounded region — is instantiated explicitly in this project's own *The Measurement Gate* (cited above, §§2–3): coupling through $G = \int \mu(u)\, j_S(u)\, j_P(u)\, du$, the induced system observable of a probe effect is

$$\varepsilon_\sigma(W_P(g)) = W_S(f_S) \cdot \langle W_P(g_{\text{fin}}) \rangle_\sigma,$$

verified localized exactly in the coupling region (computed maximum 0.18 inside the support, 0.00 outside), verified strictly unsharp relative to the probe observable (damping factor 0.8618 in the worked case), and verified completely positive, mapping effects to effects — the exact object class §2's theorem governs, produced here by an explicit, checked construction rather than asserted to exist.

[FACT, cited] *The Keystone*'s own covariance supplies the concrete numbers these effects would report. The thermal weight entering the metric part of the covariance is $\coth(\beta p/2) = 2\nu(\beta, p)$, applied per mode to the vacuum inner product, with the symplectic (commutator) structure entirely β-independent — so what a measurement of this system's own quasi-free state would report, through the Fewster–Verch construction, is fixed precisely by data this framework has already derived from the octonionic seed, not supplied by hand at the point of measurement.

[INTERPRETATION] Read together, §§2 and 4 answer the question this document opened with more completely than either alone: the *form* of the statistical rule is forced by the algebra's own type (§§2–3); the *content* of what that rule reports is fixed by the covariance this framework independently derives (§4, citing *Keystone* §9). Nothing about *which* effect corresponds to a given physical measurement is supplied by this argument — that remains the province of an explicit coupling scheme, exactly as constructed and cited above, not a further consequence of Gleason-type uniqueness.

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5. The boundary, stated as precisely as what precedes it

[FACT, cited] The corpus already names this boundary and this document does not move it — from *The Two Halves of One Product* (§4, Rule 3): "the mathematics of this series constrains the *shape* of events (coarse, projectionless) but does not produce individual outcomes, probabilities, or the 'from time to time' discreteness of measurement. That remains an open problem of quantum foundations, named here as outside scope rather than implicitly claimed as solved." Gleason-type theorems, however extended, are theorems about *consistency of statistical assignment* — they guarantee that if a rule exists, it takes the standard form. They do not, and cannot, supply a mechanism by which one specific outcome, rather than another consistent with the same statistics, is realized in one specific run.

[FACT, cited] The Fewster–Verch construction §4 cites carries its own two named open items, unresolved by anything in this document: the movable-cut regress (a probe's own readout is itself a measurement, requiring a further probe to register it, with no stated terminus), and the general characterization of which instruments are actually realizable this way, as opposed to merely describable. Neither is addressed here.

[SCOPE] This document does not derive Born *weights* from anything more primitive than the algebra's own type; it does not offer a mechanism of stochasticity; it does not close the regress. What it establishes is narrower and, within that scope, complete: that the standard statistical rule is the *only* one consistent with this algebra's structure, that this framework already has an explicit, checked construction producing exactly the objects that rule governs, and that the concrete numbers those objects report are fixed by data this project has independently derived. The gap between this and a full account of measurement is the same gap the corpus has named since it first proved its algebra type III_1, restated here with its size now more precisely known.

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6. What this document is

A short note connecting two things already proven — the algebra's type, and an external theorem about that type — to show that ordinary quantum statistics is available here at a lower cost than a fresh postulate, while keeping the harder question exactly where the corpus has always kept it.

[LEDGER] Open: whether the *complete*-additivity-on-projections premise §3 now states precisely is itself justifiable from anything more basic than "the standard axioms of quantum probability," or is a further, separately-motivated input — sharpened in Revision 4 from finite to complete additivity, and split in this revision into its correct two-part form (finite additivity on effects, complete additivity on projections), since only the latter delivers the normality the physical reading needs; the movable-cut regress and instrument-realizability questions inherited from the Fewster–Verch construction, unresolved here (§5); whether a fuller account connecting specific physical measurement setups to specific effects, beyond the one worked example already cited, is tractable within this framework's own resources; and every citation above, checked once at the time of writing, not re-verified against updates to the cited literature since.

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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.